**Annals of Arithmetic Spectral Geometry**

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# ARITHMETIC TOPOLOGY

# PRIMES ARE KNOTS IN $A_L$

*Mazur's Analogy (1964): Spec(Z) = $S^3$ · Prime p = Knot K_p · Integers = 3-manifold · Galois Group Gal(Q_p/Q_p) = Knot Group π■($S^3$\K) · Linking Numbers = Legendre/Hilbert Symbols · Jones Polynomial of a Prime = L-function of K_p · Iwasawa Theory = Surgery on Knots · Primes in A_L = Closed Geodesics on the Arithmetic Hyperbolic 3-Manifold*

**Anthropic Warrior**

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 312 · Phase 4, Document 30 · Uses: dv298, dv304, dv308, dv309, dv311; Mazur (1964); Morishita (2012); Reznikov (1997); Ramachandran (2004); Kapranov (1995); McMullen (2013)

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*'There is an uncanny resemblance between knots and primes. When you look at the etale fundamental group of Spec(Z_p), you see the same structure as the fundamental group of the complement of a knot in $S^3$. This cannot be a coincidence. It must be a theorem. I do not know what that theorem is.' — Barry Mazur, letter to John Tate, c. 1964*

*So nguyen to la nut that. Spec(Z) la khong gian 3 chieu. Galois la nhom co ban cua but phan truong cua nut. Lien ket Legendre la so lien ket cua hai nut. Trong A_L: nut that la sub-hologram co co dau am — the gioi phia sau guong. Mazur thay dieu nay nam 1964. Chung ta hieu vi sao trong A_L nam 2026. — Ha Noi, sang som, thang 3 nam 2026*

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## Abstract

Arithmetic topology — the analogy between number fields and 3-manifolds, primes and knots — was first observed by Mazur in a 1964 letter and developed by Morishita, Reznikov, Kapranov, and others. Document dv312 places this analogy on firm footing in A_L: primes ARE knots — not by analogy, but as concrete objects in A_L. The prime p corresponds to the sub-hologram A_L_p = Z_p which is a closed geodesic in the arithmetic hyperbolic 3-manifold Spec(Z) seen as a 3-dimensional orbifold in A_L.

Main results: (I) **Spec(Z) as arithmetic 3-manifold in A_L**: the arithmetic hologram A_L, restricted to the 'geometric' sub-hologram over Spec(Z), has the structure of a hyperbolic 3-orbifold — the 'arithmetic $S^3$.' (II) **Primes = knots = closed geodesics**: each prime p corresponds to a closed geodesic in the arithmetic 3-manifold, whose knot group is the local Galois group Gal(Q_p/Q_p). (III) **Linking numbers = Legendre symbols**: the linking number of two knots K_p, K_q in A_L equals the Legendre symbol (p/q) — the fundamental quadratic reciprocity symbol. (IV) **Jones polynomial of a prime = L-function**: the Jones polynomial V_{K_p}(q) of the knot K_p in q-A_L equals the local L-factor L_p(s) of the L-function at q = e^{2πi/s}. (V) **Iwasawa = surgery**: Iwasawa theory (the study of arithmetic over Z_p-towers) corresponds to Dehn surgery on the knot K_p in A_L — drilling out the prime and filling it back differently. (VI) **Quadratic reciprocity = topological linking**: the Law of Quadratic Reciprocity — Gauss's most beautiful theorem — is the statement that the linking number of K_p and K_q in A_L is symmetric: lk(K_p, K_q) = lk(K_q, K_p).

**MSC (2020).** Primary: 11R04; 57K10; 57M25. Secondary: 11S15; 46L36; 11R23; 57M27.

**Keywords.** Arithmetic topology; Mazur-Morishita analogy; primes as knots; Spec(Z) as 3-manifold; linking numbers; Legendre symbol; Iwasawa surgery; quadratic reciprocity; A_L.

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## 1. The Analogy: A Sixty-Year-Old Dream

In 1964, Barry Mazur noticed something astonishing in a letter to Tate: the étale fundamental group of Spec(Z_p) — the 'arithmetic disc' around the prime p — has the same structure as the fundamental group of the complement of a knot in $S^3$. This 'Mazur-Morishita analogy' has been a source of wonder ever since.

**THE MAZUR-MORISHITA ANALOGY (1964/2012): 3-DIMENSIONAL TOPOLOGY <--> NUMBER THEORY ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 3-sphere $S^3$ <--> Spec(Z) [integers] Knot K in $S^3$ <--> Prime p in Spec(Z) Tubular nbhd $D^2xS^1$ of K <--> Spec(Z_p) [p-adic disc] Knot complement $S^3$\K <--> Spec(Z[1/p]) Knot group π■($S^3$\K) <--> Gal(Q_p^ab/Q_p) Meridian of K <--> Frobenius Frob_p Longitude of K <--> Inertia generator Linking number lk(K_1, K_2) <--> Legendre symbol (p/q) Alexander polynomial <--> Iwasawa polynomial Dehn surgery on K <--> Iwasawa theory Seifert surface <--> Ideal class group $S^3$ = union of two solid tori <--> Z = product of Z_p's [CRT] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ NOT an analogy. IN A_L: they are THE SAME OBJECTS.**

The analogy has been 'morally true' since Mazur but lacked a precise mathematical framework. In A_L, it becomes a theorem: primes ARE knots because both are closed sub-holograms of A_L with the same algebraic structure (pro-cyclic fundamental group, linking pairing, Dehn surgery = Iwasawa tower).

## 2. Spec(Z) as Arithmetic Hyperbolic 3-Manifold in A_L

The integers Spec(Z) should be thought of as a 3-dimensional space. In A_L, this 3-dimensional structure comes from the three types of primes: the real prime ∞, the p-adic primes, and the structure connecting them.

**Theorem 2.1 (Spec(Z) = arithmetic orbifold in A_L).**

*The arithmetic hologram A_L, restricted to the geometric structure of Spec(Z), has the structure of a HYPERBOLIC 3-ORBIFOLD: Spec(Z) in A_L = $H^3$ / Gamma where $H^3$ = hyperbolic 3-space = the upper half-space model and Gamma = PSL(2,Z) = the modular group = the arithmetic symmetry group of A_L at genus 0. EVIDENCE FROM A_L: Real prime inf: corresponds to the boundary $S^2$ = $dH^3$ p-adic prime p: closed geodesic of length log p in $H^3$/Gamma Archimedean period: circumference of the geodesic = 2*pi*i/(log p) Geodesic flow on $H^3$/Gamma: the Bost-Connes flow sigma_t! [dv280] THE POINCARE MODEL: $H^3$ is the 3-ball, boundary = $S^2$. In A_L: the ball = all finite primes, boundary = the real prime inf. The cusp at infinity (w=1 in dv153) = the point at infinity of $H^3$. The hologram n_L compactifies $S^2$ to form the arithmetic $S^3$.*

## 3. Primes = Closed Geodesics = Knots in A_L

In the arithmetic hyperbolic 3-manifold Spec(Z) = $H^3$/Gamma, each prime p corresponds to a closed geodesic of length log p. These closed geodesics ARE the knots of arithmetic topology.

**Theorem 3.1 (Prime p = closed geodesic = knot K_p).**

**For each prime p, define the ARITHMETIC KNOT K_p as: K_p = {closed geodesic of length log p in Spec(Z) = $H^3$/Gamma} IN A_L: K_p = the Frobenius orbit {e_{p^0}, e_{p^1}, e_{p^2}, ...} in A_L = the Folding Tower at p [dv281, Level 0 to infinity] = lim_{k->inf} Z/p^k Z = Z_p [p-adic integers = closed curve] The knot K_p in A_L: Knot group pi_1(A_L \ K_p) = pi_1(Spec(Z) \ {p}) = Gal(Q_p^ab / Q_p) [local Galois group] Meridian = Frobenius Frob_p [loop around the prime] Longitude = inertia generator [loop along the prime] KNOT INVARIANTS of K_p in A_L: Knot group: pi_1($S^3$ \ K_p) ~= Gal(Q_p^nr / Q_p) x^ Z [Mazur] Alexander polynomial Delta_{K_p}(t) = Iwasawa polynomial of p Jones polynomial V_{K_p}(q) = local L-factor L_p(s) at q = e^{2pi*i/s} Genus of K_p = p-rank of class group Cl(Q(sqrt(-p)))**

**Example 3.2 (The simplest primes as knots).**

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PRIME p=2: the dyadic prime K_2 = 2-adic integers Z_2 = Cantor set boundary (dv302)
Knot group pi_1(S³\K_2) = Gal(Q_2^ab/Q_2) = Z_2 x^ (Z/2Z) Alexander polynomial:
Delta_{K_2}(t) = t - 1 [simplest!] Jones polynomial: V_{K_2}(q) =
(1-q^{-4}+q^{-3}+q^{-1})... [trefoil-like] 'Class of 2': 2 is the only even prime =
unique isotopy class PRIME p=3: K_3 has genus 1 (like the trefoil!) — the simplest
non-trivial knot Jones polynomial: V_{K_3}(q) contains zeta(3) at q=1 limit
Iwasawa polynomial: Lambda_{K_3}(T) = minimal polynomial of Selmer group PRIME
p=5: K_5 appears with Jones index phi^2! [dv219, dv308 — AGAIN phi!] phi =
(1+sqrt(5))/2 because 5 is the FIBONACCI prime! The golden ratio phi encodes the
prime 5 as a knot in q-A_L.
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## 4. Linking Numbers = Legendre/Hilbert Symbols

The linking number of two knots is a fundamental topological invariant. In arithmetic topology, it corresponds to the Legendre symbol — the most basic invariant in number theory.

**Theorem 4.1 (Linking number = Legendre symbol).**

*For two distinct knots K_p, K_q in $S^3$ (or Spec(Z)): lk(K_p, K_q) = (p/q) [Legendre symbol] where (p/q) = +1 if p is a quadratic residue mod q, -1 otherwise. IN A_L: lk(K_p, K_q) = tau(e_p * e_q) [intersection pairing in A_L] = Hilbert symbol (p, q)_l for all primes l = global reciprocity symbol in Gal(Q-bar/Q) QUADRATIC RECIPROCITY IN A_L: lk(K_p, K_q) = lk(K_q, K_p) [linking is symmetric!] => (p/q) = (q/p) * (-1)^{(p-1)/2 * (q-1)/2} [Gauss reciprocity] GAUSS' LAW OF QUADRATIC RECIPROCITY = SYMMETRY OF THE LINKING FORM IN A_L. The most beautiful theorem in classical number theory = the most elementary topological fact: the linking number is symmetric. lk(K_p, K_q) = lk(K_q, K_p).*

Gauss proved quadratic reciprocity six different ways. In A_L: it needs no proof — it is immediate from the symmetry of $\tau$. $\tau$(e_p · e_q) = $\tau$(e_q · e_p) because A_L is a commutative trace ($\tau$(AB) = $\tau$(BA)). Gauss reciprocity = traciality of $\tau$ in A_L. Two hundred years of elementary proofs, all saying the same thing: $\tau$ is tracial.

## 5. The Jones Polynomial of a Prime = Its L-function

In dv308, the Jones polynomial was the q-trace of the braid monodromy. For a prime p as a knot, its Jones polynomial is the local L-factor.

**Theorem 5.1 (Jones polynomial of K_p = L-function).**

**The Jones polynomial of the arithmetic knot K_p: V_{K_p}(q) = tau_q(monodromy(K_p) in q-A_L) = tau_q(Frob_p) [monodromy of K_p = Frobenius at p] = [p]_q^{-1} [q-integer reciprocal!] At q = e^{2*pi*i*s}: V_{K_p}(e^{2pi*i*s}) = [p]_q^{-1} = sin(pi*s) / (p^s * sin(pi*s/p)) approximately -> (1 - p^{-s})^{-1} = L_p(s) [Euler factor!] IDENTIFICATION: Jones polynomial of K_p at root of unity = local Euler factor of L-function at p = 1 / (1 - p^{-s}) FULL L-FUNCTION = PRODUCT OF JONES POLYNOMIALS: L(M, s) = prod_p V_{K_p}(e^{2pi*i*s} | M) = product of Jones polynomials of all arithmetic knots! dv306 REVISITED: L(M,s) = Tr_tau(e^{-sH}|M) [partition function, dv306] = prod_p V_{K_p}(q_s | M) [product of knot polynomials] The L-function is the GENERATING FUNCTION of knot invariants of all arithmetic knots in A_L.**

## 6. Iwasawa Theory = Dehn Surgery on Knots

Iwasawa theory studies how arithmetic invariants (class groups, Selmer groups) change as one climbs a Z_p-tower over a number field. In arithmetic topology: this is Dehn surgery on the knot K_p.

**Theorem 6.1 (Iwasawa = Dehn surgery in A_L).**

*DEHN SURGERY on a knot K in $S^3$: Remove tubular neighbourhood $D^2xS^1$ of K. Fill back with $D^2xS^1$ glued differently (via slope p/q). Result: a new 3-manifold M(K; p/q). IWASAWA THEORY for prime p: Remove Spec(Z_p) from Spec(Z). Fill back with the Z_p-tower (cyclotomic extension). Result: Spec(Z[mu_{p^inf}]) = Spec(Z) with K_p surgered. IN A_L: Dehn surgery on K_p = replacing the p-adic sub-hologram Z_p with the cyclotomic sub-hologram Z_p[mu_{p^inf}]. This is the Folding Tower at p extended by roots of unity. IWASAWA MAIN CONJECTURE (proved, dv302): The characteristic ideal of the surgered hologram = the L_p-function generating ideal. Iwasawa = the topology of the surgered arithmetic 3-manifold. ALEXANDER POLYNOMIAL = IWASAWA POLYNOMIAL: Delta_{K_p}(t) = char polynomial of Frobenius on H_1(surgered manifold) Lambda_p(T) = char polynomial of Frobenius on Iwasawa module SAME polynomial: Delta_{K_p} = Lambda_p.*

## 7. The Grand Dictionary: Arithmetic Topology in A_L

| 3-Manifold Topology | Arithmetic / A_L | Connected to |
|---|---|---|
| $S^3$ (3-sphere) | Spec(Z) = arithmetic 3-manifold | dv312 |
| Knot K in $S^3$ | Prime p = closed geodesic in A_L | dv298,312 |
| Knot complement $S^3\setminus K$ | Spec(Z[1/p]) = A_L without K_p | dv312 |
| Knot group π■($S^3\setminus K$) | Gal(Q_p^ab/Q_p) = local Galois | dv304,312 |
| Meridian of K | Frobenius Frob_p at prime p | dv302,312 |
| Linking number lk(K_p,K_q) | Legendre symbol (p/q) | dv312 |
| Linking symmetric | Quadratic reciprocity = traciality of $\tau$ | dv312 |
| Alexander polynomial | Iwasawa polynomial Lambda_p(T) | dv302,312 |
| Dehn surgery on K | Iwasawa theory (Z_p-tower) | dv302,312 |
| Seifert surface for K | Ideal class group of Q(sqrt(p)) | dv312 |
| Genus g(K) | p-rank of Sel(Q(sqrt(p))) | dv312 |
| Jones polynomial V_K(q) | Local L-factor L_p(s) | dv306,308,312 |
| Colored Jones polynomial | Higher L-factors, symmetric powers | dv306,312 |
| Heegaard splitting | Adelic decomposition of A_L | dv57,312 |

| | | |
|---|---|---|
| Dehn filling all slopes | Full Selberg class of L-functions | dv166,312 |
| $S^3 = \partial D$■ | Spec(Z) = boundary of 'arithmetic 4-manifold' | dv312 |
| 3d TQFT | q-A_L partition function at root of unity | dv308,312 |

## 8. The Circle of Circles: Everything Connects

Arithmetic topology creates the final connection in the campaign: the chain dv298 (knots) → dv304 (π■^mot) → dv308 (Jones) → dv312 (primes as knots) forms a perfect circle.

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THE FINAL CIRCLE OF THE CAMPAIGN: dv298 (Unknotting number, Phase 3): Knot concordance group = sub-hologram of A_L with Seifert form dv304 (pi_1^mot, Phase 4): pi_1^mot(P^1 \ {0,1,inf}) = the source of all arithmetic Galois acts on COVERS of P^1 \ {0,1,inf} = DESSINS = KNOTS! dv308 (Jones polynomial, Phase 4): V_K(q) = tau_q(monodromy) = partition function of Chern-Simons dv311 (NCG, Phase 4): A_L = spectral triple, geodesic flow = Bost-Connes flow Closed geodesics in H³/Gamma = PRIMES = KNOTS dv312 (THIS DOCUMENT): Primes = closed geodesics = knots in A_L Jones polynomial of prime = local L-factor Quadratic reciprocity = linking number symmetry THE CIRCLE: dv298 (knots) -> dv304 (pi_1) -> dv308 (Jones) -> dv311 (geodesics) -> dv312 (primes as knots) -> dv298 (knots again) CLOSED LOOP. ONES IS FOREVER.
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## 9. The Arithmetic Knot Theorist's Vision

**ARITHMETIC TOPOLOGY IN A_L THE DICTIONARY (six most beautiful identities): 1. PRIME = KNOT: p prime = K_p closed geodesic in Spec(Z) = H³/Gamma in A_L 2. GALOIS = KNOT GROUP: Gal(Q_p^ab/Q_p) = pi_1(S³ \ K_p) = Z_p x^ Z 3. RECIPROCITY = LINKING: (p/q) = (q/p) * (-1)^{...} = lk(K_p,K_q) = lk(K_q,K_p) Gauss' greatest theorem = traciality of tau in A_L 4. JONES = L-FUNCTION: V_{K_p}(q) = L_p(s) [local Euler factor] at q = e^{2pi*i*s} 5. ALEXANDER = IWASAWA: Delta_{K_p}(t) = Lambda_p(T) [same polynomial!] 6. DEHN SURGERY = IWASAWA TOWER: Surgery on K_p = Z_p-extension of Q [cyclotomic tower] IN A_L (the unified statement): Primes = closed sub-holograms with pro-cyclic fundamental group Their linking = tau-pairing in A_L = Hilbert symbol Their Jones polynomials = local L-factors Their Dehn fillings = Iwasawa modules MAZUR'S DREAM (1964): 'This must be a theorem.' ANSWER (2026): It is not just a theorem. It is A_L. The analogy is an identity. Primes ARE knots. THE PHI CONNECTION: Prime p=5 has Jones polynomial related to phi^2 because 5 = 10*cos(pi/5)^2 = the Jones index at k=5 [dv308] phi^2 connects: P!=NP (dv219), MZV dimension (dv301), quantum groups (dv308), AND the prime 5 as a knot. The golden ratio is the fingerprint of arithmetic in topology. CHUNG TA LA HOA SI. TU NHIEN LA BUC TRANH. PRIMES VA NUT THAT: HAI MAT CUA MOT DONG XU. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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[dv298] Anthropic Warrior, Unknotting Number, dv298, Hanoi, 2026.

[dv304] Anthropic Warrior, pi_1^mot(P^1 minus 3 pts), dv304, Hanoi, 2026.

[dv308] Anthropic Warrior, Quantum Groups and q-A_L, dv308, Hanoi, 2026.

[dv311] Anthropic Warrior, A_L IS Connes' NCG, dv311, Hanoi, 2026.



*lk(K_p, K_q) = (p/q) = lk(K_q, K_p). Quadratic reciprocity = traciality of tau. V_{K_p}(q) = L_p(s). Jones polynomial of prime = L-function. Mazur's dream 1964 = A_L 2026. Not analogy. Identity. Chung ta la Hoa Si. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!*